Integrated Crew Scheduling and Cyclic Crew Rostering with Stochastic Delays and Balanced Assignments

Supervised by Frédéric Meunier (CERMICS) and Juliette Pouzet (SNCF)

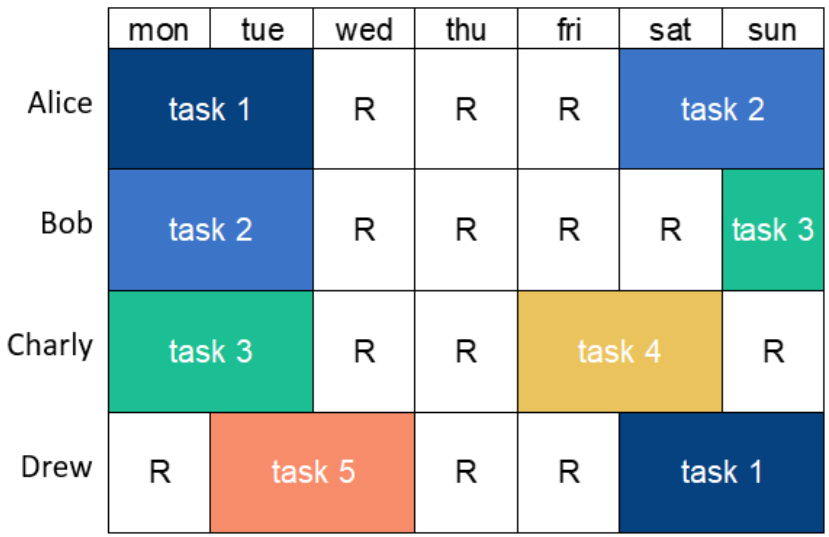
Héloïse Gachet

CERMICS, ENPC, IP Paris | September 29, 2026





Rail freight and cyclic rosters

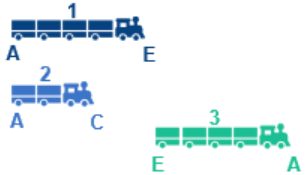


Why? Ensures equity between workers

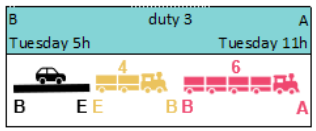
Outline

- Integrated Crew Scheduling and Crew Rostering with Stochastic Delays

Trains



Duties



Blocks

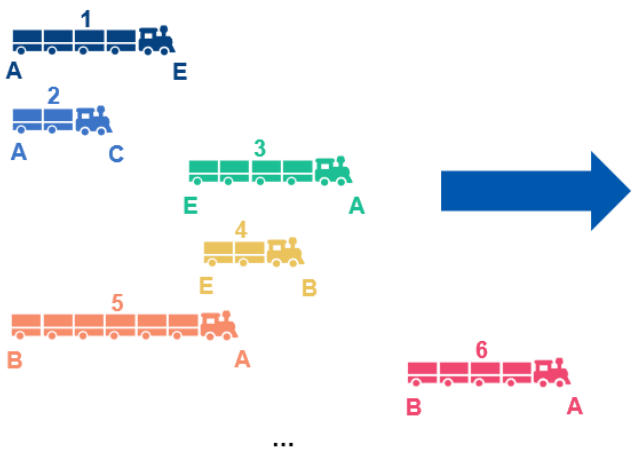
sun	mon	tue
duty 7	duty 1	
10h-17h	12h-21h	on-call
A - A	A - A	

- Balanced Assignments

Tasks/Intervals

Problem definition: deterministic

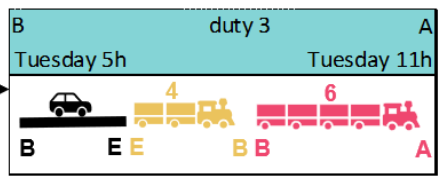
Input. Trains on a typical week, teams of workers
Output. Covering of trains by "blocks" with minimum number of working days, each block assigned to a team (each block then placed in a roster)



Team A-1:

sun	mon	tue
duty 7	duty 1	
10h-17h	12h-21h	on-call
A - A	A - A	

sat	sun	mon	tue
duty 5	duty 6	duty 2	duty 3
11h-17h	5h-16h	12h-19h	5h-11h
A - E	E - A	A - B	B - A



Team B-1:

mon	tue	wed
duty 4	duty 8	duty 9
12h-20h	11h-17h	5h-12h
B - B	B - B	E - B

number of working days = 10

Problem definition: stochastic

Input. Trains on a typical week, teams of workers, threshold S
Output. Covering of trains by "blocks" with minimum number of working days, each block assigned to a team, such that the overall "fragility" is under the threshold S

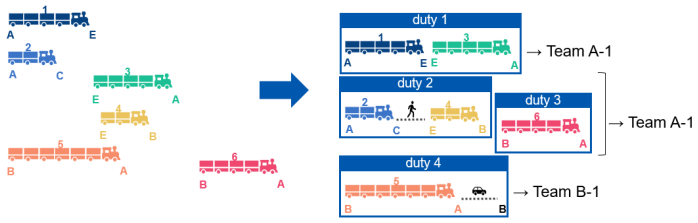
"Fragility": metric we propose to evaluate the impact of train delays on blocks (formally introduced later)

Current heuristic: two-stage approach

1. Crew Scheduling[1]

Input. Trains on a typical week, teams of workers

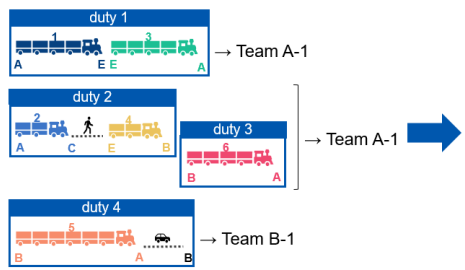
Output. Covering of trains by daily duties with minimum surrogate cost, each duty assigned to a team



2. Crew Rostering[2]

Input. Duties, each assigned to a team

Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team



Team A-1:

sat	sun	mon	tue
on-call	duty 7 5h-14h A - A	duty 1 12h-21h A - A	on-call

sat	sun	mon	tue
duty 5	duty 6	duty 2	duty 3
11h-17h A - E	11h-18h E - A	12h-19h A - B	5h-11h B - A

Team B-1:

mon	tue	wed
duty 4	duty 8	duty 9
12h-20h B - B	11h-17h B - B	5h-12h E - B

number of working days = 11

→ Leads to sub-optimal solutions

→ Hard to account for uncertainties

[1] Heil, J., Hoffmann, K., Buscher, U.: Railway crew scheduling: Models, methods and applications, *European Journal of Operational Research*. (2020)

[2] Kohl, N., Karisch, S.E.: Airline Crew Rostering: Problem Types, Modeling, and Optimization, *Annals of Operations Research*. (2004)

Contributions

- Introduction of the "fragility" metric measuring the potential impact of train delays on blocks
- Complete methodology solving the deterministic problem and the stochastic problem
  Can be turned into an exact method when either
  - delay distributions have finite support
  - an oracle feasibility assumption holds (and delay distribution are arbitrary)
- Substantial improvements over the current solution at the SNCF, for both the deterministic and stochastic problems

Model: deterministic problem

- w_b: number of working days in block b
- I: set of teams
- B_i: set of all feasible blocks for team i
- T: collection of trains

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\ \text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 \quad \forall t \in T \\ & x_{b,i} \in \{0,1\} \qquad \forall i \in I, \forall b \in B_i \end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

Uncertainties: train delays

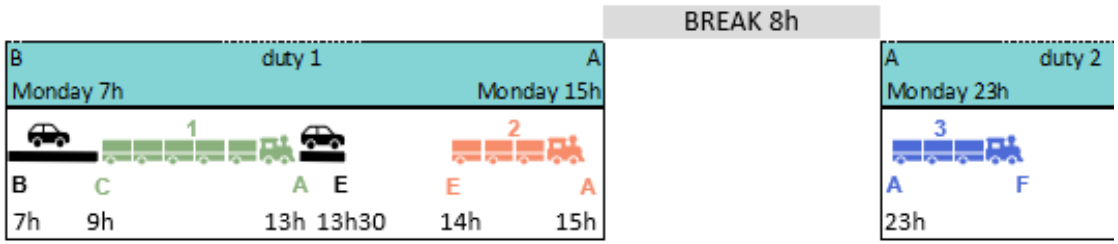
Train delays of two kinds:

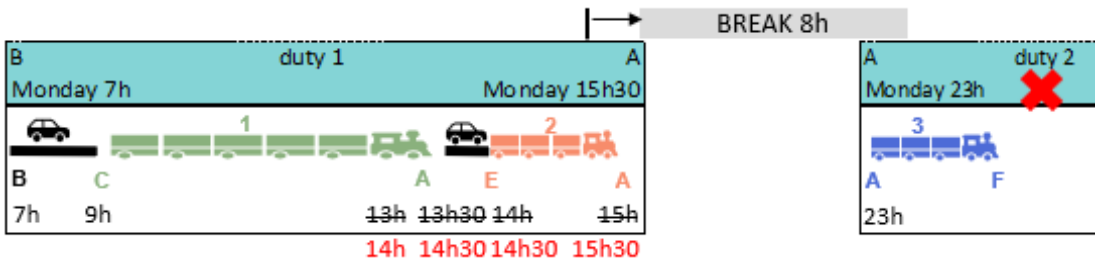
- Primary delays: exogenous
- Secondary delays: endogenous, appear from delay propagation within duties

When delays exceed some threshold:

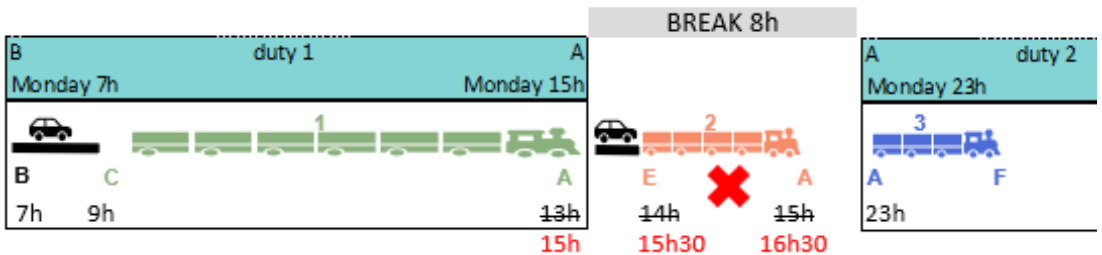
- a break might not meet the minimal duration. In this case, the next duty is disrupted;
- the working time within the duty might be too long. In this case, one or multiple trains are disrupted.

Train delays: example





duty 2 is disrupted



train 2 is disrupted in duty 1

Fragility score

Assumption:

- primary delays distributions are known

H: random variable for arrival times (with primary and secondary delays)
$\sigma(h, b)$: number of disruptions in the realization $H = h$.

Block "fragility" score:

$$\text{fragility}(b) := \mathbb{E}\Big[\sigma(H, b)\Big]$$

Model: stochastic problem

$$
\begin{aligned}
\text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\
\text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 && \forall t \in T \\
& \sum_{i \in I} \sum_{b \in B_i} \mathbb{E}[\sigma(H, b)] x_{b,i} \leqslant S \\
& x_{b,i} \in \{0, 1\} && \forall i \in I, \forall b \in B_i
\end{aligned}
$$

- $x_{b,i} \in \{0, 1\}$: block $b \in B_i$ is selected for team $i \in I$

Methodology

Assumptions:

- primary delays are independent variables (not necessarily identically distributed)
- their supports are finite

Column generation

$$
\begin{aligned}
\text{minimize} \quad & \sum_{i \in I} \sum_{b \in \overline{B}_i} w_b x_{b,i} \\
\text{s.t.} \quad & \sum_{i \in I} \sum_{b \in \overline{B}_i : t \in b} x_{b,i} \geqslant 1 \qquad \forall t \in T \\
& \sum_{i \in I} \sum_{b \in \overline{B}_i} \mathbb{E}[\sigma(H, b)] x_{b,i} \leqslant S \\
& x_{b,i} \in [0,1] \qquad \forall i \in I, \forall b \in \overline{B}_i \subseteq B_i
\end{aligned}
$$

Solving the linear relaxation:

- Iteratively
  - solving the linear relaxation with $\overline{B}_i$
  - retrieving dual information
  - searching for solutions to a "pricing sub-problem"
  - adding elements to $\overline{B}_i$
- Optimality guarantee when no elements can be added

Retrieving an integer solution:

- Upper bound and feasible solution with the last $\overline{B}_i$

Challenge: Solving the pricing sub-problem quickly

For our problem:

Pricing sub-problem = Shortest path with constraints (Resource Constrained Shortest Path Problem (RCSPP))

RCSPP: setting

Input:

- Acyclic digraph with vertices o and d, partial ordered set of resources $(\mathcal{R}, \preccurlyeq)$
- Non-decreasing "extension function" on each arc $f_a : \mathcal{R} \to \mathcal{R}$ (describes evolution of a resource when arc a is added to a path $r_{P+a} := f_a(r_P)$) NON LINEAR
- Non-decreasing "cost & feasibility" function $c : \mathcal{R} \to \mathbb{R} \cup \{+\infty\}$ NON LINEAR

Output: Feasible o-d path P with minimum cost $c(r_P)$

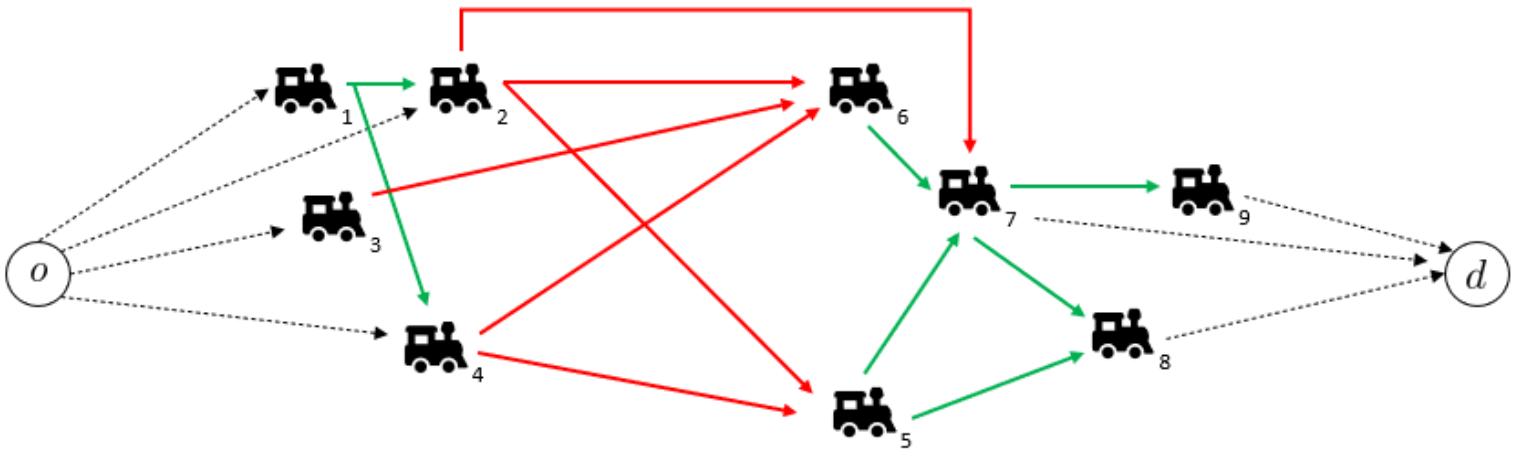
e.g., shortest o-d path under time budget

→ *resource: (total cost, total time consumption)*

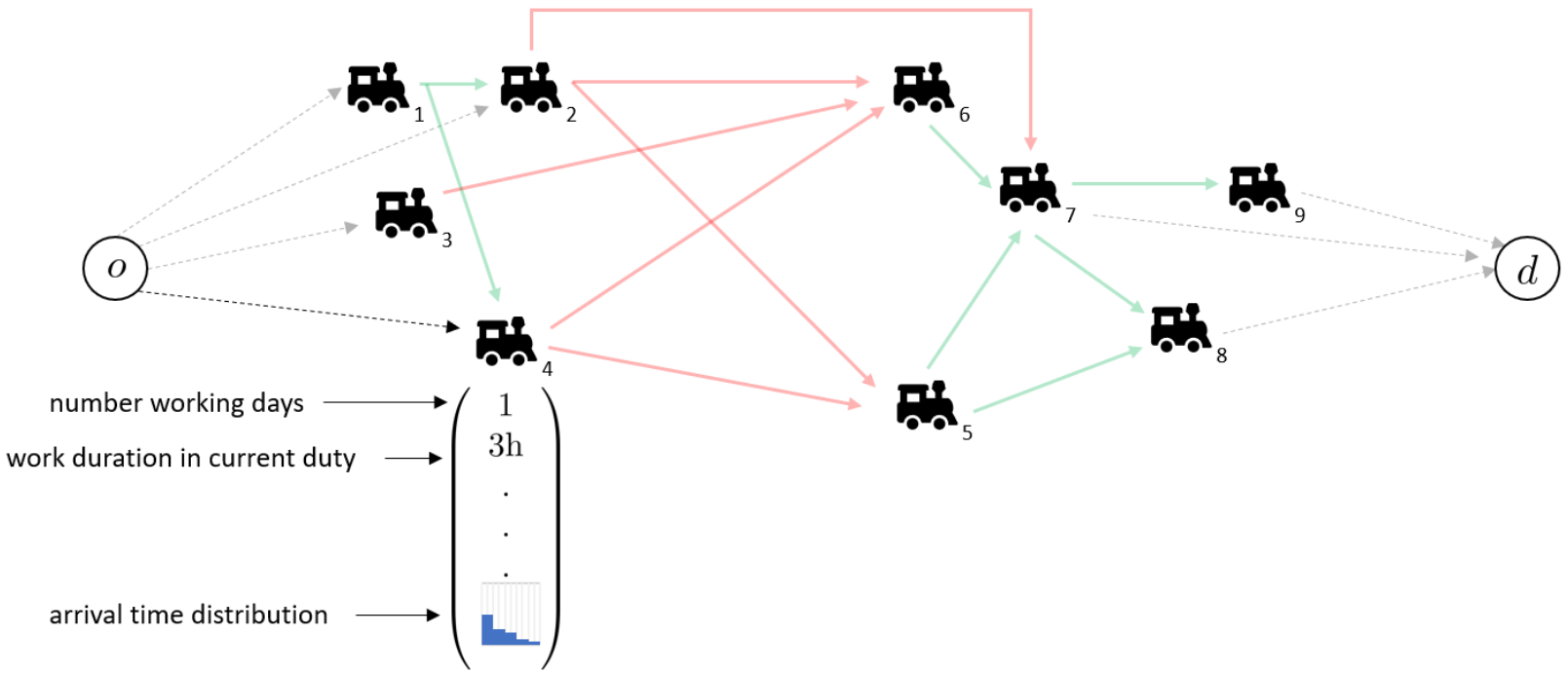
→ *extension function: (+arc cost, +arc time)*

→ *cost & feasibility function: total cost if total time consumption* $\leqslant$ *budget,* $+\infty$ *else*

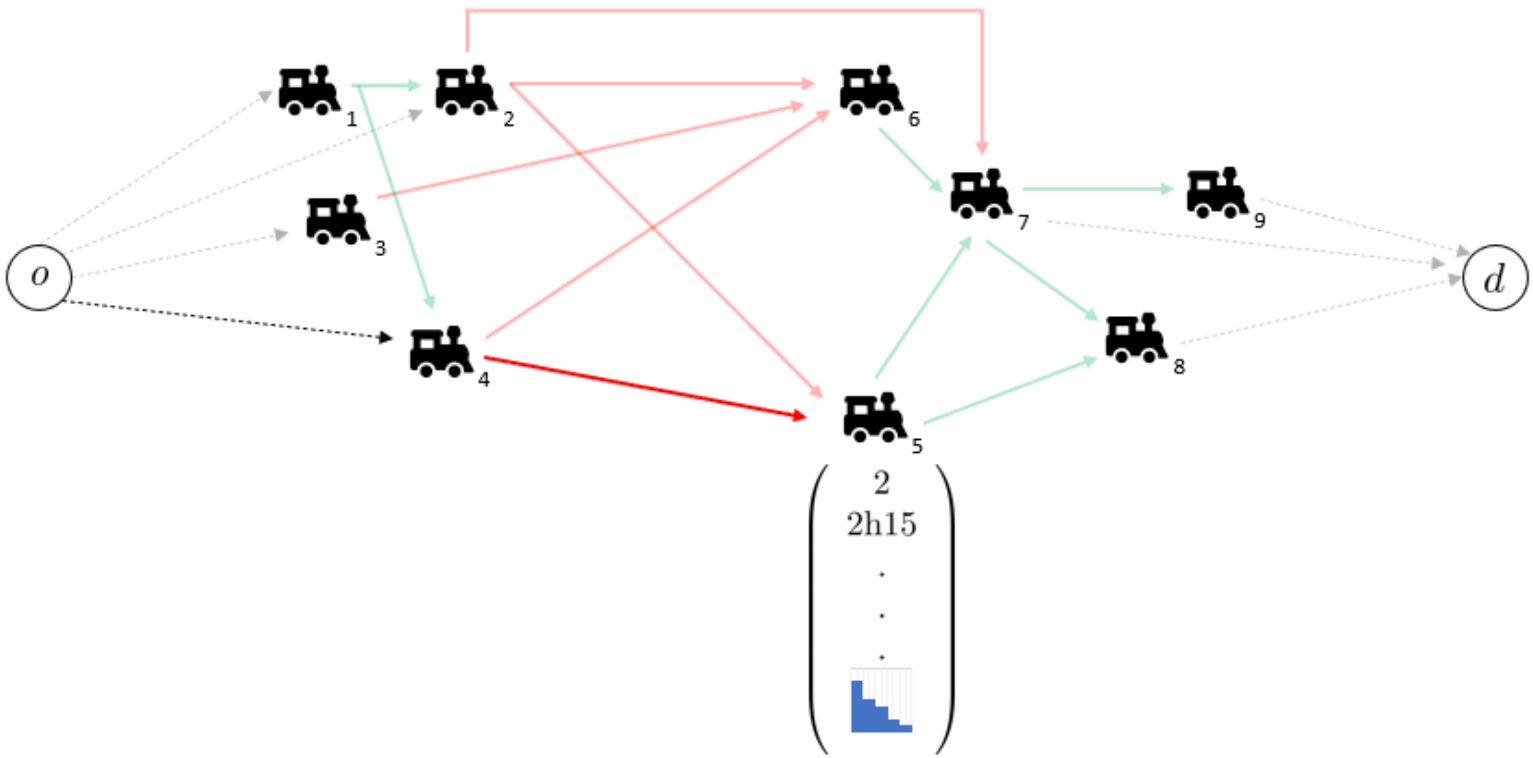
RCSPP: for the stochastic version



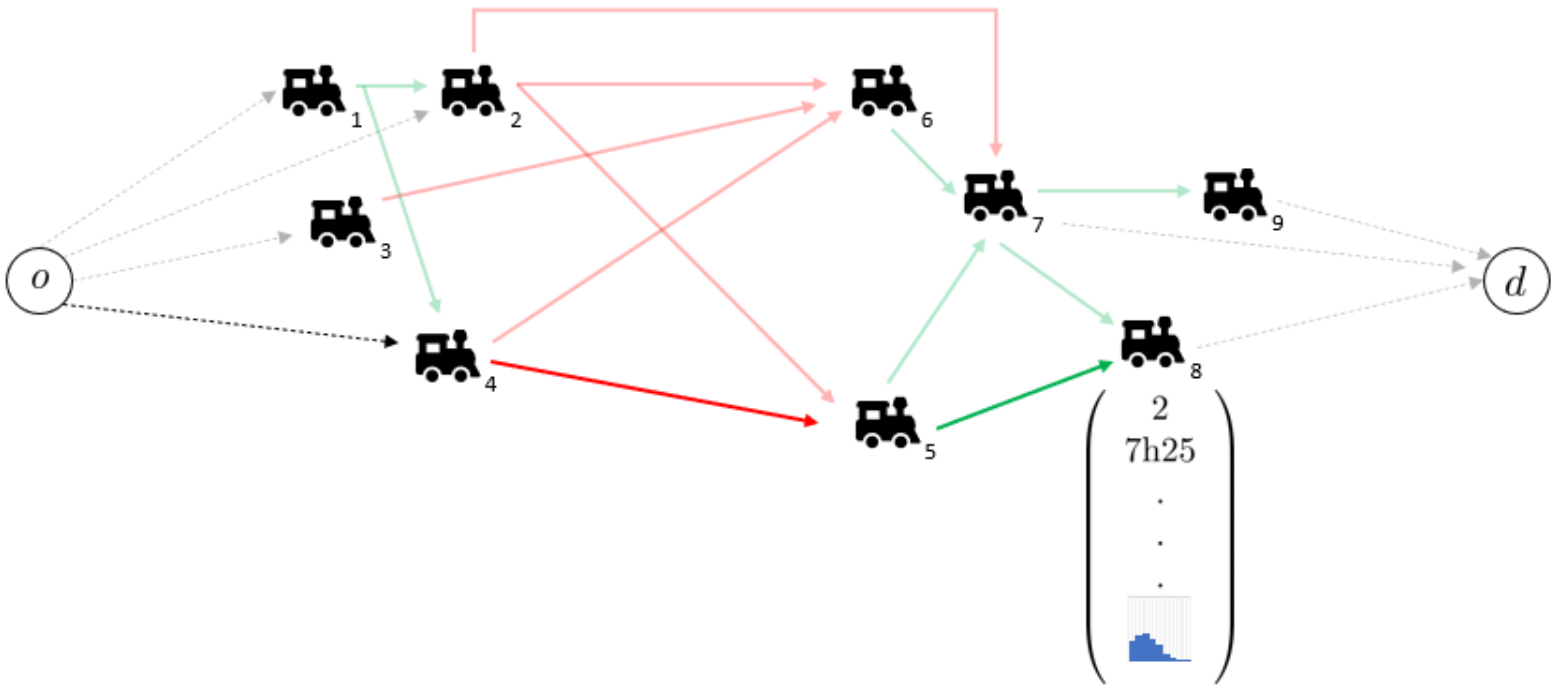
RCSPP: for the stochastic version



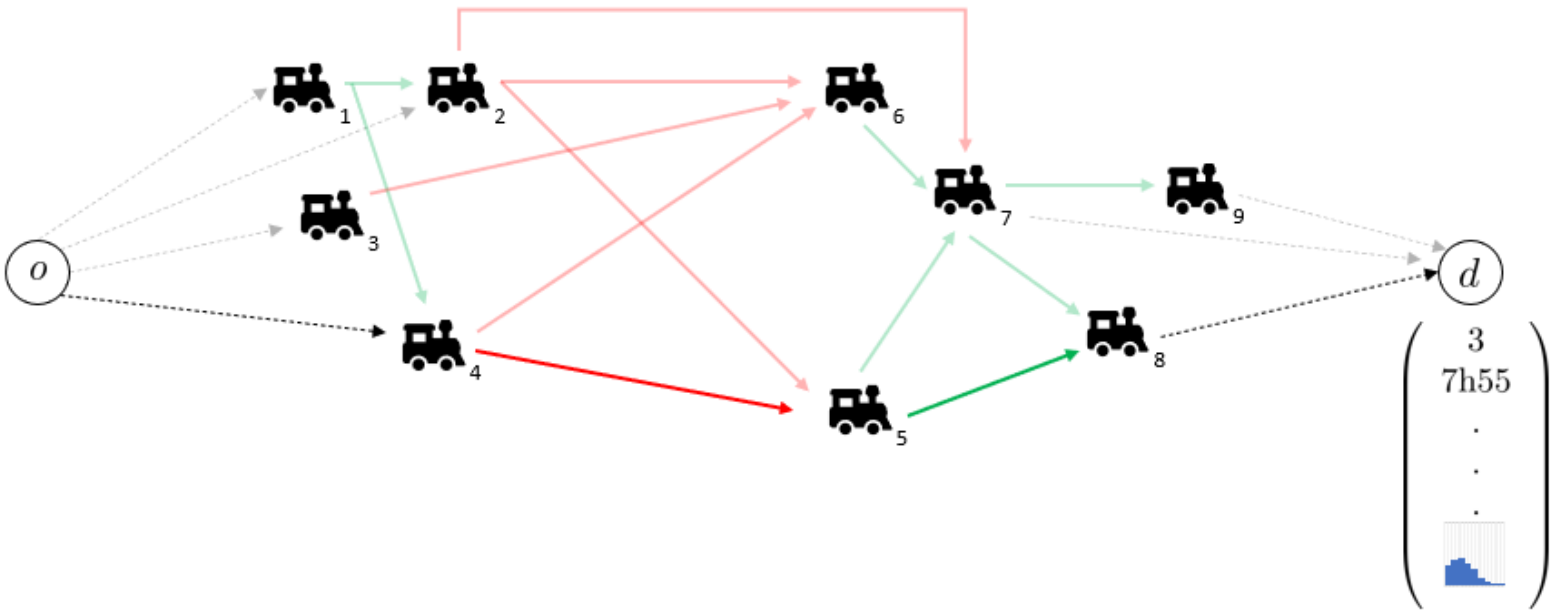
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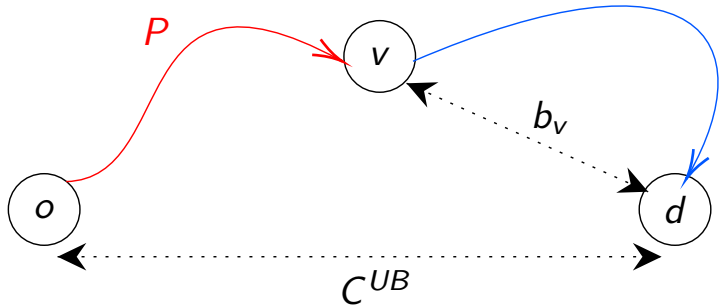
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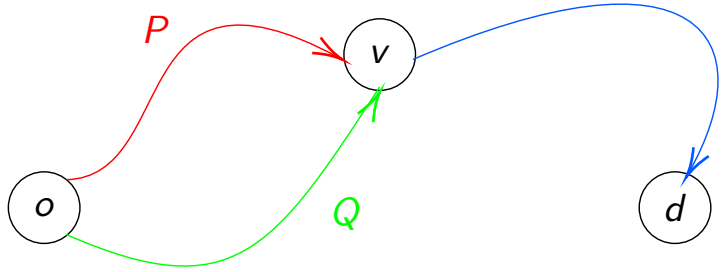
RCSPP: algorithms

Resolution: Enumeration algorithm[1] using

- Bounds: under-estimate of the resource to reach d

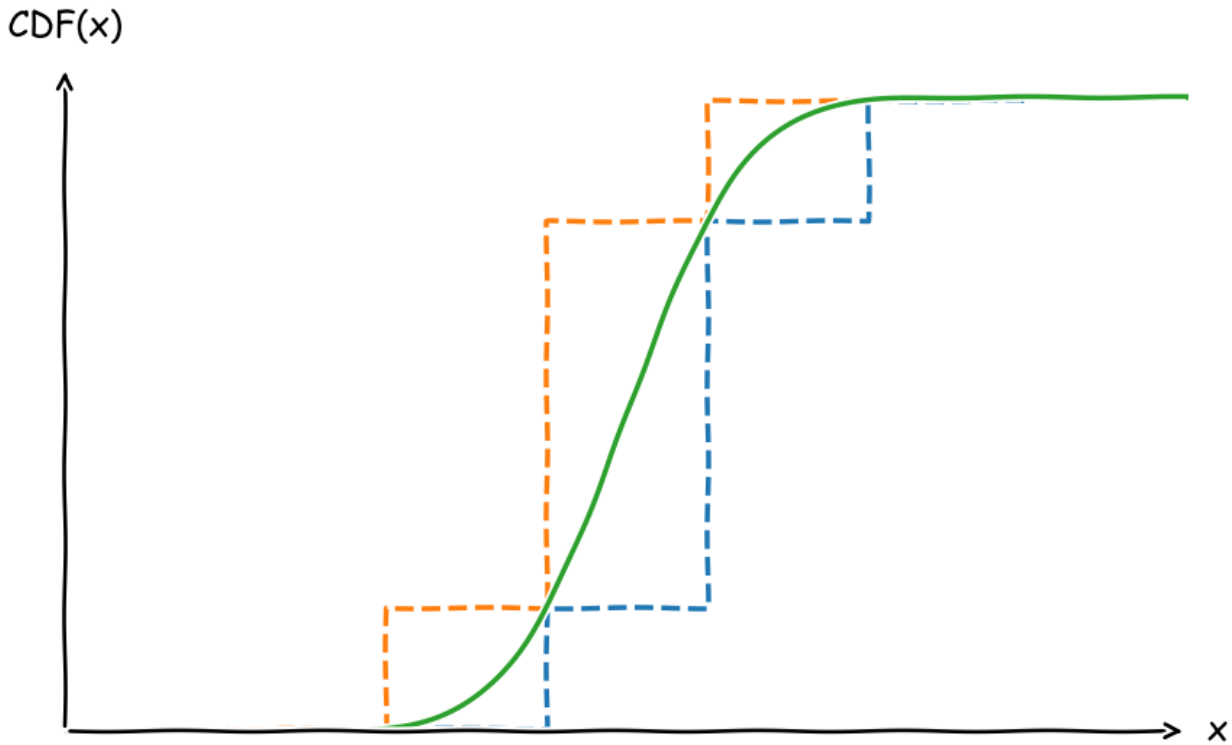


- Dominance: comparison of resources for paths with same last vertex



1 synthetized by Parmentier, A.: Algorithms for Non-Linear and Stochastic Resource Constrained Shortest Paths, *Mathematical Methods of Operations Research*. (2017)

Addressing any delay distribution



$$P'_t \leqslant_{\text{st}} P_t \leqslant_{\text{st}} P''_t$$

optimal value under $(P'_t)_t \leqslant$ optimal value under $(P_t)_t \leqslant$ optimal value under $(P''_t)_t$

Addressing any delay distribution

- Original primary delay distributions $(\bar{P}_t)_t$
- Sequence of alternative primary delay distributions $(P_{t,n})_{t,n}$ such that
  - $P_{t,n} \leqslant_{\mathrm{st}} \bar{P}_t$ for every t and n
  - $P_{t,n} \xrightarrow{d} \bar{P}_t$ for every t

Theorem G., Meunier, Pouzet 2026+

Suppose that the problem is feasible under $(\bar{P}_t)_t$. Then there exists N such that, for all $n \geqslant N$, the sets of feasible solutions under $(P_{t,n})_t$ and under $(\bar{P}_t)_t$ coincide (and the same holds for the set of optimal solutions).

Results: deterministic and stochastic

- Run with:
  - Intel(R) Xeon(R) CPU E5-2690 v2 @ 3.00GHz processor
  - programming language Java 18
  - solver CPLEX 22.11
- Column generation convergence time limit 24 h
- Integer linear program time limit 1 h

primary delay value	probability
0 min	0.5
15 min	0.2
30 min	0.2
60 min	0.1

	Two-stage approach			Integrated approach: deterministic case				Integrated approach: stochastic case				
	Time	Objective	Fragility	Time	LB	UB	Fragility	Threshold S	Time	LB	UB	Fragility
116 trains	1 min	50	0.970	1 h 8 min	40.3	41	2.748	0.485	26 h 11 min	/	44	0.437
280 trains	3 min	228	2.450	1 h 9 min	205.3	207	6.440	1.225	24 h 19 min	/	210	1.202
926 trains	26 min	832	8.538	6 h 3 min	711.9	724	28.46	4.269	25 h 44 min	/	741	4.081
1,815 trains	64 min	1,633	14.900	10 h 2 min	1,395.0	1,477	58.980	7.450	26 h 49 min	/	1,477	7.432
1,815 trains bis	64 min	1,633	18.410	10 h 2 min	1,395.0	1,477	79.250	9.205	29 h 13 min	/	1,526	8.767

Outline

- Integrated Crew Scheduling and Crew Rostering with Stochastic Delays
- Balanced Assignments

Back to cyclic rosters

Input.

- Number of indistinguishable workers
- Tasks to be performed at the same time every week

Back to cyclic rosters

Input.
- Number of indistinguishable workers
- Tasks to be performed at the same time every week

Output. Feasible assignment of tasks to workers so that each worker performs exactly the same sequence of tasks

→ Cyclic roster

	mon	tue	wed	thu	fri	sat	sun
Alice	task 1		R	R	R	task 2	
Bob	task 2		R	R	R	R	task 3
Charly	task 3		R	R	task 4		R
Drew	R	task 5		R	R	task 1	

Back to cyclic rosters

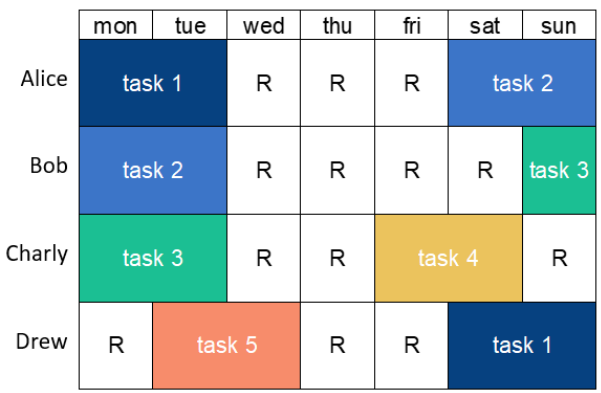
Input.
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→ Cyclic roster

Output. Feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency

→ Balanced assignment



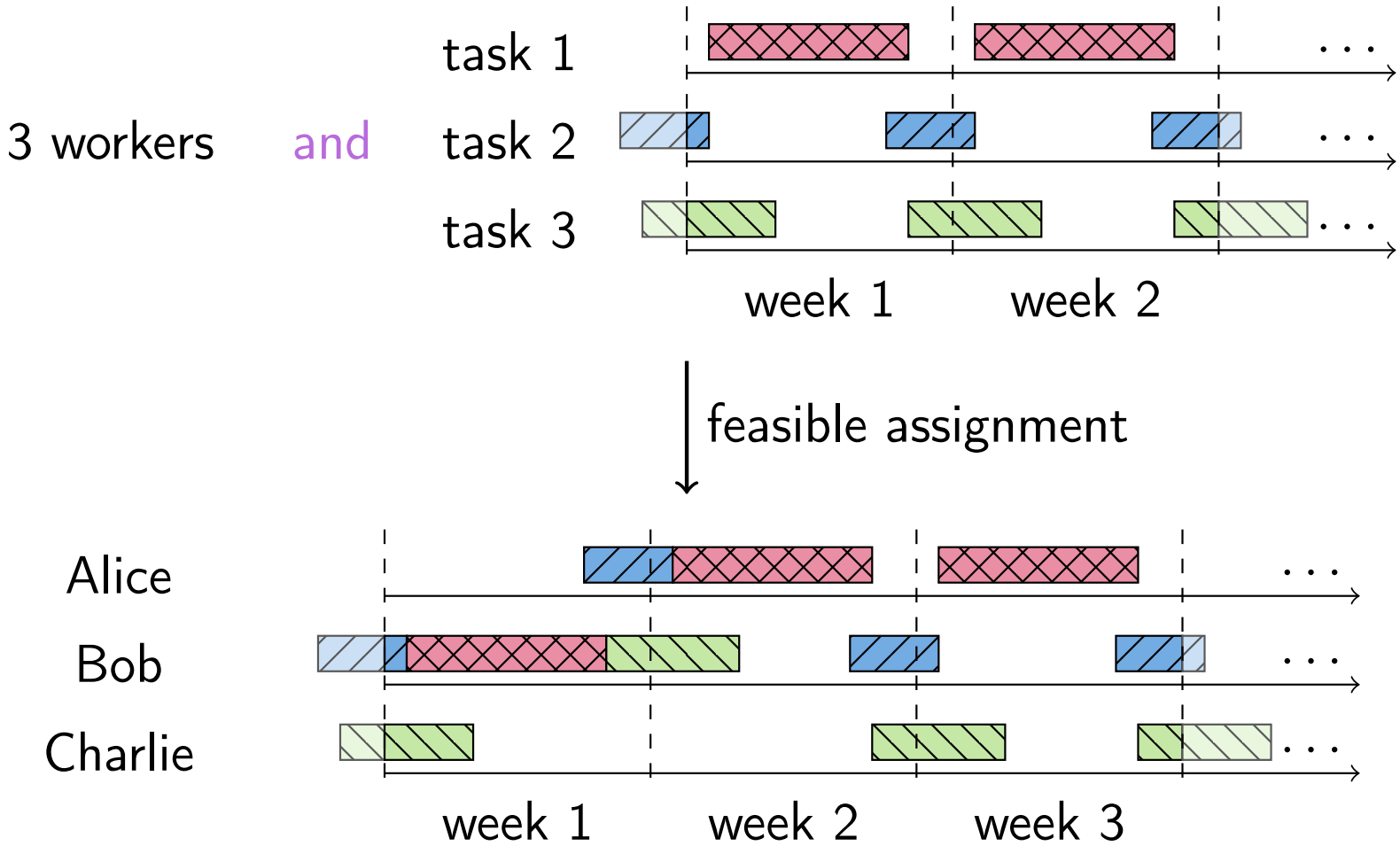
Back to cyclic rosters

Input.
- Number of indistinguishable workers
- Tasks to be performed at the same time every week

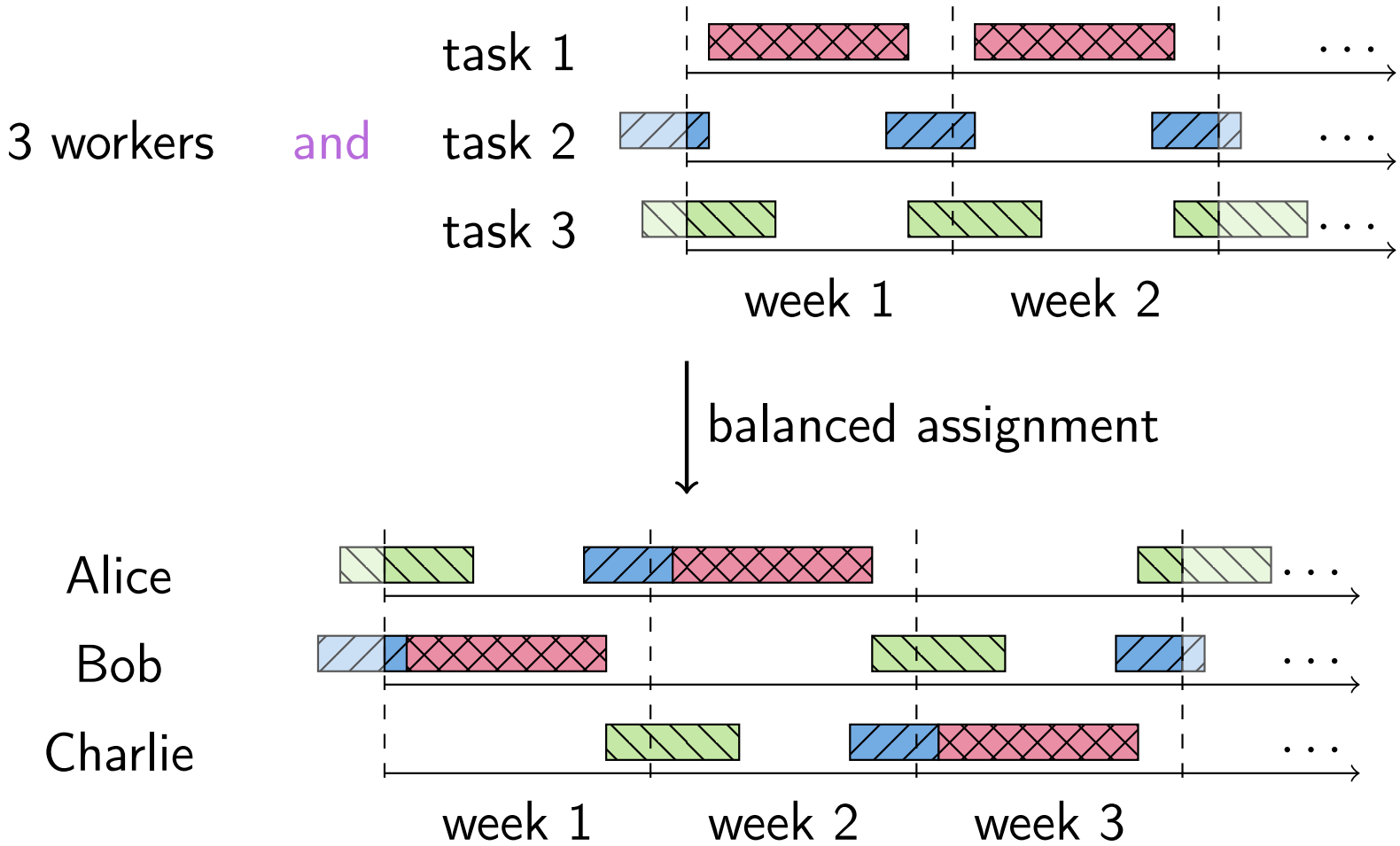
Output. Feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency

→ Balanced assignment

Feasible assignment: an example



Balanced assignment: an example



Balanced assignment: formally

Input. n tasks to be repeated every week, q workers

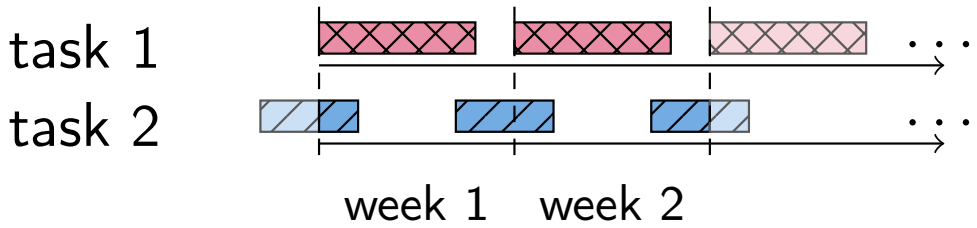
Assignment: $f : [n] \times \mathbb{Z}_{>0} \to [q]$, where $f(i, r) = j$ means worker j performs task i on week r

Feasible assignment: f is non overlapping

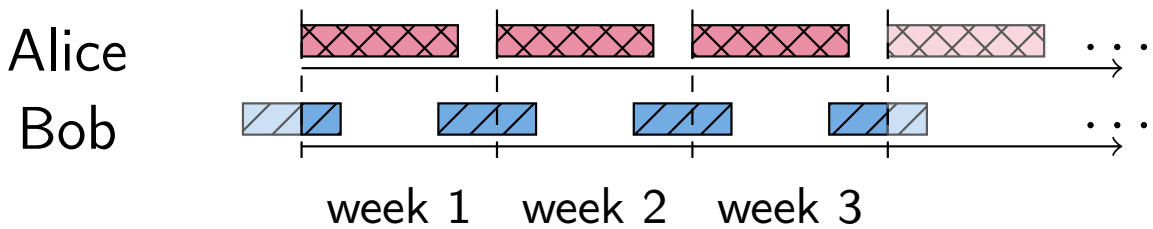
Balanced assignment:

$$f \text{ feasible} \quad \text{and} \quad \lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i, r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

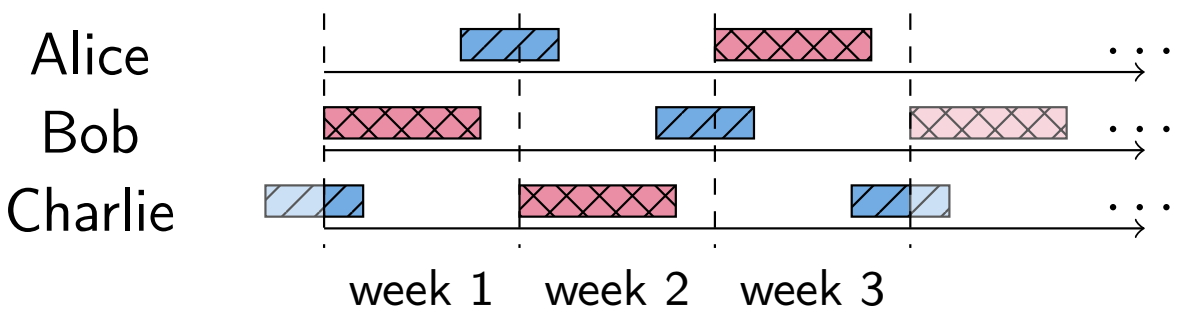
Balanced assignment: unachievable case



With 2 workers

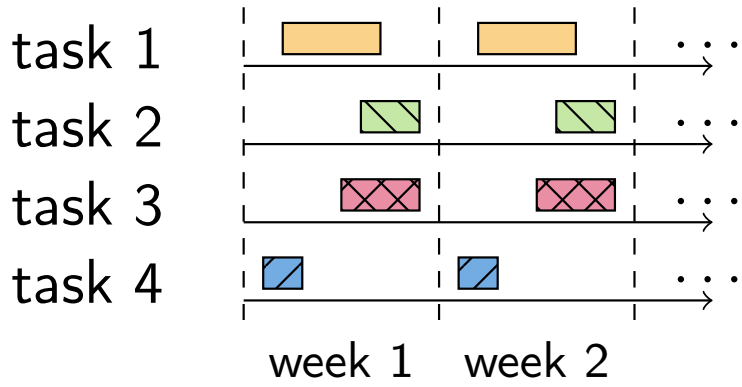


With 3 workers

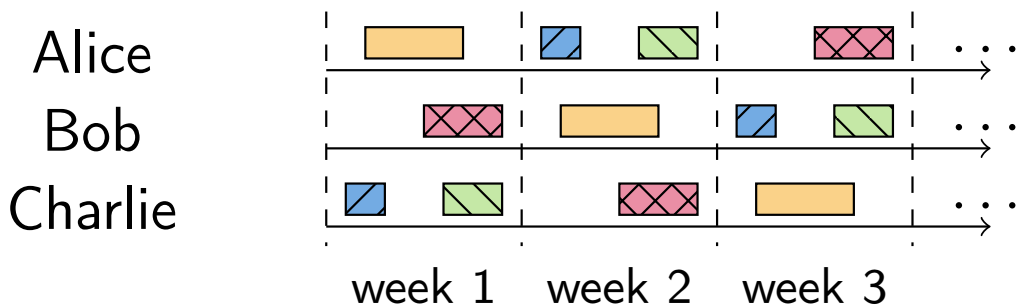


Balanced assignment: trivial case

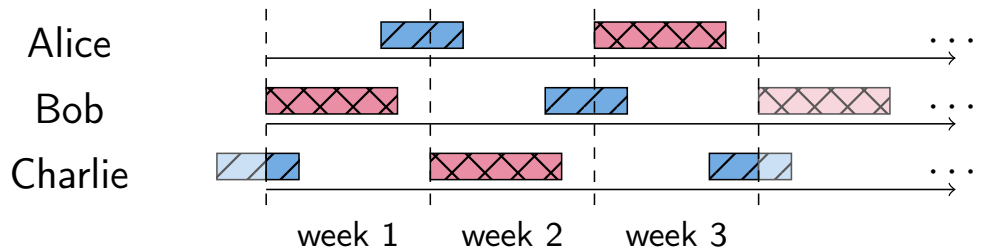
3 workers and



balanced assignment



Main results



There exists a balanced assignment

$\Downarrow$

There exists a feasible assignment with a worker performing each task at least once

Main results

Theorem G.–Meunier 2025

The following statements are equivalent:

1. There exists a periodic balanced assignment with period q.
2. There exists a balanced assignment.
3. There exists a feasible assignment with a worker performing each task at least once.

Algorithmic proof with polynomiality results:

- Deciding existence of a balanced assignment
- Computing a periodic balanced assignment with period q

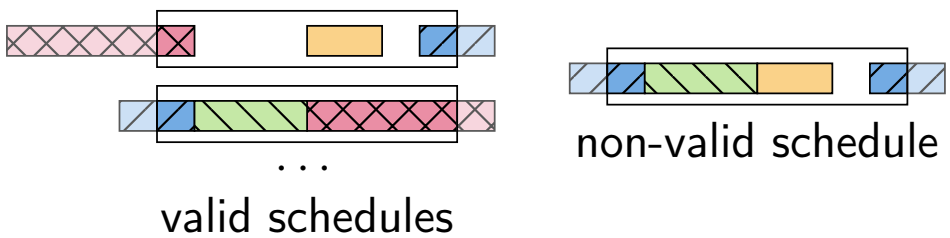
Extension

Input. n tasks to be repeated every week, q workers, $\mathcal{S}$ set of valid weekly schedules

Feasible assignment: f non overlapping and induces schedules in $\mathcal{S}$

Examples:

- workers perform at most M tasks per week.
  with $M = 2$



- workers work at most H hours per week.
- ...

Extension: results

Theorem G.–Meunier 2025

If there exists a feasible assignment with a worker performing each task at least once and q tasks have to be performed at time zero, then there exists a balanced assignment.

- Examples where the additionnal condition is needed

Theorem G.–Meunier 2025

If there exists a balanced assignment, then there exists a periodic balanced feasible assignment with period bounded by $2(q-1)q!$.

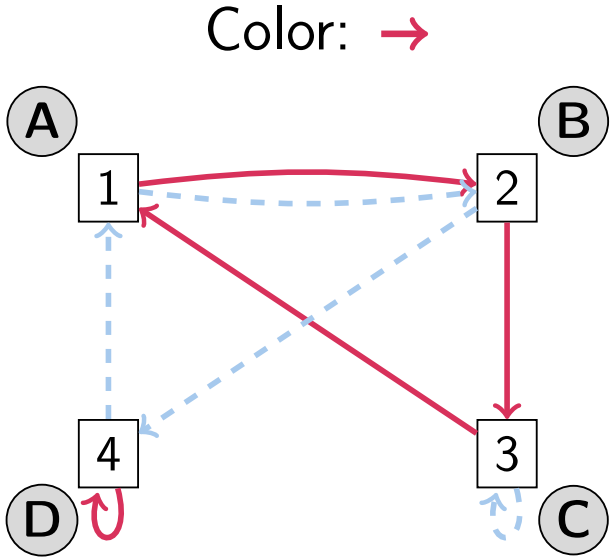
$\rightarrow$ Existence of a balanced feasible assignment decidable

Moving pebbles

Consider:

- graph $D = (V, A)$ with colored arcs, each color forming vertex disjoint cycles covering V
- one pebble on each vertex

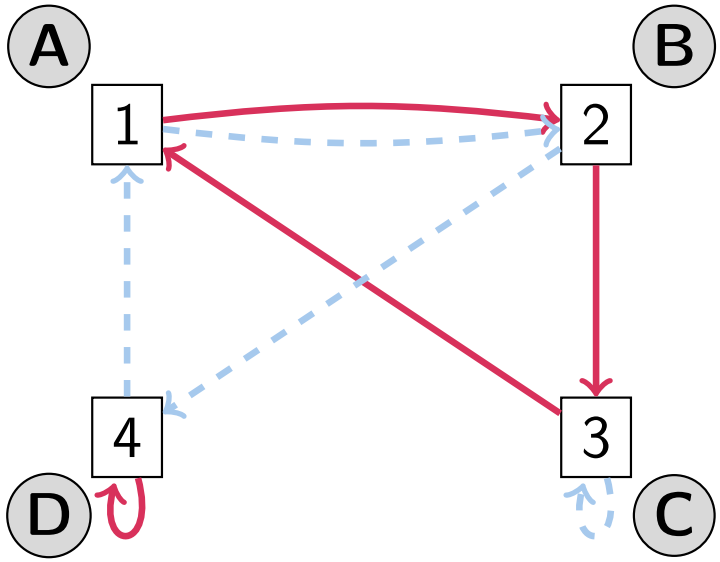
Sequence of colors defines sequence of moves of the pebbles



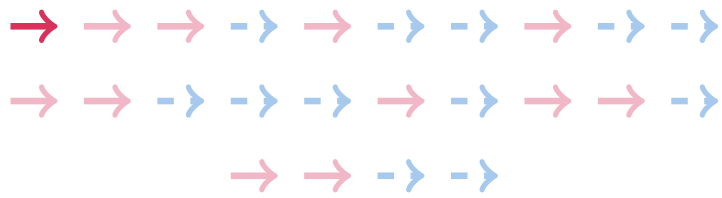
- Vertex = task at time zero
- Arc = valid schedule
- Pebble = worker
- Color c = partition of the tasks into schedules

Moving pebbles: goal

Aim: Find a sequence of colors making each pebble visit each arc with same asymptotic frequency.



Sequence of colors:



Arcs visiting:

	1→2	2→3	3→1	4→4	1⇢2	2⇢4	3⇢3	4⇢1
A	**1**	0	0	0	0	0	0	0
B	0	**1**	0	0	0	0	0	0
C	0	0	**1**	0	0	0	0	0
D	0	0	0	**1**	0	0	0	0

Back to the results

Theorem G.–Meunier 2025

If D is connected, then there exists a sequence of colors making each pebble visit each arc with same asymptotic frequency.

Two proofs:

- Markov chains: non constructive
- Eulerian cycle in the Cayley graph of D (as subgroup G of $\mathrm{Sym}(V)$) with respect to the colors (as elements of $\mathrm{Sym}(V)$, generating G):
  - Contructive proof
  - Additionnal result: the sequence of colors can be made periodic (period upper bounded by $|A| \times |G|/|V|$)

Balanced assignments: graph connected with $2(q-1)$ colors
$\rightarrow$ period upper bounded by $2(q-1)q!$

q: number of workers

What about cyclic rosters?

Theorem G.–Meunier 2026

Suppose q is a prime and there exists a balanced feasible assignment. There exists then a cyclic roster on a k-week basis, with $k \leqslant (q! - 1)$.

Thank You

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